XGB_tuning

October 31, 2025

[2]:
```python
from sklearn.metrics import r2_score
import numpy as np
import pandas as pd
from sklearn.metrics import mean_absolute_error,mean_squared_error
import optuna          # for tunning
import xgboost as xgb
from sklearn.model_selection import KFold
import os
from sklearn.preprocessing import StandardScaler
```

[29]:
```python
X = pd.read_csv("X_train.csv")
y = pd.read_csv("y_train.csv").squeeze()
X_test = pd.read_csv("X_test.csv")
y_test = pd.read_csv("y_test.csv").squeeze()

print("Train-test data loaded successfully!")
print(f"X_train: {X.shape}, X_test: {X_test.shape}")
```

```
Train-test data loaded successfully!
X_train: (780, 2), X_test: (195, 2)
```

[16]:
```python
X
```

[16]:

	temp	salt
0	24.511438	35.126246
1	22.507472	34.661103
2	25.007761	34.412981
3	24.268401	33.891595
4	25.901783	34.010018
..	…	…
775	25.677783	32.765154
776	23.029611	34.186860
777	27.189343	34.418573
778	23.569874	32.369443
779	24.592546	34.197649

[780 rows x 2 columns]

[17]:
```
# sc = StandardScaler()
# X = sc.fit_transform(X)
# X_test = sc.transform(X_test)
```

[18]:
```
kfold = KFold(n_splits=10)
```

[19]:
```
def objective(trial):
    scores = []
    for train_ix, val_ix in kfold.split(X):
        param = {
        'lambda': trial.suggest_float('lambda', 0.2, 1.0),
        'alpha': trial.suggest_float('alpha', 0.2, 1),
        'subsample': trial.suggest_float('subsample', 0.2, 1.0),
        'colsample_bytree': trial.suggest_float('colsample_bytree', 0.2,1.0),
        'max_depth' : trial.suggest_int('max_depth', 1, 20, step=1),
        'min_child_weight' : trial.suggest_int('min_child_weight', 1, 20),
        'learning_rate': trial.suggest_float('learning_rate',0.1,0.3),
       #      "booster": trial.suggest_categorical("booster", ["gbtree",␣
 ↪"dart"]),
       #  'reg_alpha': trial.suggest_float('reg_alpha', 0.0, 2.0),
       #  'reg_lambda': trial.suggest_float('reg_lambda', 0.0, 2.0),
       # "grow_policy": trial.suggest_categorical("grow_policy", ["depthwise",␣
 ↪"lossguide"]),
        'gamma' : trial.suggest_float('gamma', 0, 1),
        'n_estimators': trial.suggest_int('n_estimators',0,500,step=50),
        'random_state' : 0, 'n_jobs':-1}
#         if param["booster"] == "dart":
#                 param["sample_type"] = trial.
 ↪suggest_categorical("sample_type", ["uniform", "weighted"])
#                 param["normalize_type"] = trial.
 ↪suggest_categorical("normalize_type", ["tree", "forest"])
#                 param["rate_drop"] = trial.suggest_loguniform("rate_drop",␣
 ↪1e-8, 1.0)
#                 param["skip_drop"] = trial.suggest_loguniform("skip_drop",␣
 ↪1e-8, 1.0)



        X_train, X_val = X.iloc[train_ix], X.iloc[val_ix]
        y_train, y_val = y.iloc[train_ix], y.iloc[val_ix]

        xgb_model = xgb.XGBRegressor(**param)
        xgb_model.fit(X_train, y_train, verbose=False,eval_set=[(X_val,y_val)])

        xgb_preds = xgb_model.predict(X_val)
        mse = mean_squared_error(y_val, xgb_preds,squared=False)
```

```
        #r2 = r2_score(y_val, xgb_preds)
        scores.append(mse)


        return np.mean(scores)
```

[20]:
```
study = optuna.create_study(sampler=optuna.samplers.
 ↪CmaEsSampler(),direction='minimize')
```

```
[I 2025-10-26 18:26:42,835] A new study created in memory with name: no-name-
fe2873bf-bedc-4eb9-94fa-60b6b0cf373b
```

[21]:
```
study.optimize(objective, n_trials=50,n_jobs=-1,show_progress_bar=True)
```

```
  0%|          | 0/50 [00:00<?, ?it/s]
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
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/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
```

```
[I 2025-10-26 18:26:43,231] Trial 4 finished with value: 5.71440748130664 and
parameters: {'lambda': 0.37956552075160943, 'alpha': 0.49316985894482307,
'subsample': 0.6815795348197182, 'colsample_bytree': 0.9666575058173634,
'max_depth': 16, 'min_child_weight': 2, 'learning_rate': 0.14580005059640136,
'gamma': 0.5983849425388922, 'n_estimators': 0}. Best is trial 4 with value:
5.71440748130664.
[I 2025-10-26 18:26:43,238] Trial 6 finished with value: 5.71440748130664 and
parameters: {'lambda': 0.8474088730640095, 'alpha': 0.3264293232332025,
'subsample': 0.9647692752762993, 'colsample_bytree': 0.7512824734903949,
```

```
'max_depth': 12, 'min_child_weight': 17, 'learning_rate': 0.18507890295759016,
'gamma': 0.7248099723653314, 'n_estimators': 0}. Best is trial 4 with value:
5.71440748130664.
[I 2025-10-26 18:26:43,251] Trial 9 finished with value: 5.71440748130664 and
parameters: {'lambda': 0.7549083915968573, 'alpha': 0.7126468461691824,
'subsample': 0.2678806551633464, 'colsample_bytree': 0.3676523419567852,
'max_depth': 14, 'min_child_weight': 9, 'learning_rate': 0.12255429690602454,
'gamma': 0.961818768198573, 'n_estimators': 0}. Best is trial 4 with value:
5.71440748130664.
[I 2025-10-26 18:26:43,255] Trial 10 finished with value: 5.71440748130664 and
parameters: {'lambda': 0.7439254096084278, 'alpha': 0.8468145771294078,
'subsample': 0.3016252555788612, 'colsample_bytree': 0.9898960229514373,
'max_depth': 11, 'min_child_weight': 6, 'learning_rate': 0.2585338464031061,
'gamma': 0.584382820900826, 'n_estimators': 0}. Best is trial 4 with value:
5.71440748130664.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
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/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:47,135] Trial 16 finished with value: 1.2277459295059292 and
parameters: {'lambda': 0.6128228082611658, 'alpha': 0.5911181720075103,
'subsample': 0.4533371437501633, 'colsample_bytree': 0.23479964021417743,
'max_depth': 3, 'min_child_weight': 19, 'learning_rate': 0.2732038233220577,
'gamma': 0.905160390120468, 'n_estimators': 50}. Best is trial 16 with value:
1.2277459295059292.
[I 2025-10-26 18:26:47,188] Trial 40 finished with value: 1.2931607022341658 and
parameters: {'lambda': 0.5252109103570253, 'alpha': 0.4834496743063026,
'subsample': 0.6090854062052478, 'colsample_bytree': 0.7925291968782511,
'max_depth': 7, 'min_child_weight': 10, 'learning_rate': 0.18009575994742064,
'gamma': 0.5418059796542504, 'n_estimators': 50}. Best is trial 16 with value:
1.2277459295059292.
[I 2025-10-26 18:26:47,244] Trial 17 finished with value: 1.3429308314725283 and
parameters: {'lambda': 0.8449169151714673, 'alpha': 0.4152060323064829,
'subsample': 0.49318851333909663, 'colsample_bytree': 0.3979228801199468,
'max_depth': 9, 'min_child_weight': 1, 'learning_rate': 0.21425621622029045,
'gamma': 0.8963250933332195, 'n_estimators': 50}. Best is trial 16 with value:
```

```
1.2277459295059292.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:47,519] Trial 12 finished with value: 1.3065406315972292 and
parameters: {'lambda': 0.9965322693336736, 'alpha': 0.22812781881164154,
'subsample': 0.23231395322935652, 'colsample_bytree': 0.4418978131510911,
'max_depth': 3, 'min_child_weight': 17, 'learning_rate': 0.2695270997779985,
'gamma': 0.8389354406417039, 'n_estimators': 100}. Best is trial 16 with value:
1.2277459295059292.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:49,455] Trial 14 finished with value: 1.2227390130648852 and
parameters: {'lambda': 0.6156541051182567, 'alpha': 0.7262959435632699,
'subsample': 0.6734027691341069, 'colsample_bytree': 0.3703731641235166,
'max_depth': 1, 'min_child_weight': 19, 'learning_rate': 0.11647247779004799,
'gamma': 0.05612923343124632, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:50,436] Trial 18 finished with value: 1.3278218201724916 and
parameters: {'lambda': 0.9112191642745846, 'alpha': 0.24542651764088552,
'subsample': 0.8847629296640904, 'colsample_bytree': 0.5960503524177259,
'max_depth': 19, 'min_child_weight': 6, 'learning_rate': 0.15729391783204744,
'gamma': 0.9320319882999544, 'n_estimators': 50}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:50,442] Trial 48 finished with value: 1.2949205259577676 and
parameters: {'lambda': 0.46994543696094776, 'alpha': 0.5676803574642035,
'subsample': 0.5858216242157663, 'colsample_bytree': 0.7029777435161233,
'max_depth': 7, 'min_child_weight': 11, 'learning_rate': 0.1976216264333454,
'gamma': 0.45464169446467395, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:52,015] Trial 34 finished with value: 1.344711593243686 and
parameters: {'lambda': 0.5311769478679491, 'alpha': 0.6192150257417621,
'subsample': 0.4881622579944239, 'colsample_bytree': 0.44149121843022776,
'max_depth': 7, 'min_child_weight': 7, 'learning_rate': 0.20713941165084698,
'gamma': 0.48777253733559683, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:52,978] Trial 2 finished with value: 1.2250516799217241 and
parameters: {'lambda': 0.8616911626633439, 'alpha': 0.39381744674995683,
'subsample': 0.928479552054106, 'colsample_bytree': 0.5701506884920491,
'max_depth': 1, 'min_child_weight': 6, 'learning_rate': 0.25335855628283616,
'gamma': 0.7579295340610792, 'n_estimators': 350}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:53,651] Trial 26 finished with value: 1.3660886680233095 and
parameters: {'lambda': 0.7822044083081523, 'alpha': 0.8001662349567944,
'subsample': 0.5310743948718697, 'colsample_bytree': 0.4014291034685805,
'max_depth': 11, 'min_child_weight': 8, 'learning_rate': 0.27028806471565625,
'gamma': 0.36236339972739184, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:53,949] Trial 46 finished with value: 1.4183127712182382 and
```

```
parameters: {'lambda': 0.4881102148734635, 'alpha': 0.43146562467328886,
'subsample': 0.771092590946971, 'colsample_bytree': 0.5528758468097666,
'max_depth': 10, 'min_child_weight': 6, 'learning_rate': 0.21556098417440966,
'gamma': 0.2632469415032788, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:54,139] Trial 3 finished with value: 1.3627231497947259 and
parameters: {'lambda': 0.5561872612760175, 'alpha': 0.3708545898911371,
'subsample': 0.5145585351887374, 'colsample_bytree': 0.7101459192927044,
'max_depth': 3, 'min_child_weight': 7, 'learning_rate': 0.18584057760724454,
'gamma': 0.5249369080004359, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:54,473] Trial 33 finished with value: 1.3298028185885409 and
parameters: {'lambda': 0.7809383075813698, 'alpha': 0.4260563808254361,
'subsample': 0.622627000288857, 'colsample_bytree': 0.6411559293399638,
'max_depth': 5, 'min_child_weight': 10, 'learning_rate': 0.16754680127219046,
'gamma': 0.46009189626979274, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:56,262] Trial 35 finished with value: 1.2995694880627053 and
parameters: {'lambda': 0.5508350831379372, 'alpha': 0.6295053641719095,
'subsample': 0.4803269881318706, 'colsample_bytree': 0.7861225501883962,
'max_depth': 9, 'min_child_weight': 14, 'learning_rate': 0.1718230923754229,
'gamma': 0.35429432717809506, 'n_estimators': 150}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:57,451] Trial 5 finished with value: 1.3365214732543005 and
parameters: {'lambda': 0.35616333551819984, 'alpha': 0.5594598386471787,
```

```
'subsample': 0.3011860090609464, 'colsample_bytree': 0.8241395464829706,
'max_depth': 19, 'min_child_weight': 6, 'learning_rate': 0.16674778019673525,
'gamma': 0.05141564018733191, 'n_estimators': 150}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:57,530] Trial 37 finished with value: 1.3735299471832203 and
parameters: {'lambda': 0.7141182001901025, 'alpha': 0.6825579824543682,
'subsample': 0.688859858295853, 'colsample_bytree': 0.6633176332072124,
'max_depth': 9, 'min_child_weight': 10, 'learning_rate': 0.1677115192734917,
'gamma': 0.3708035511977026, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
```

```
[I 2025-10-26 18:26:58,096] Trial 28 finished with value: 1.4290571928003875 and
parameters: {'lambda': 0.6300675926520776, 'alpha': 0.400393312341604,
'subsample': 0.5755178575043391, 'colsample_bytree': 0.5410087505351033,
'max_depth': 16, 'min_child_weight': 6, 'learning_rate': 0.2133333758293277,
'gamma': 0.5503401348479967, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
```

```
[I 2025-10-26 18:26:58,443] Trial 49 finished with value: 1.3785666933286087 and
parameters: {'lambda': 0.5233599449553782, 'alpha': 0.6695950931672419,
'subsample': 0.6102622571655911, 'colsample_bytree': 0.38627856429244023,
'max_depth': 9, 'min_child_weight': 12, 'learning_rate': 0.20492629622982456,
'gamma': 0.4676012660710883, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:58,541] Trial 47 finished with value: 1.386770632495457 and
parameters: {'lambda': 0.6722594204106931, 'alpha': 0.6328537806484367,
'subsample': 0.8331438467595336, 'colsample_bytree': 0.676903184696396,
'max_depth': 8, 'min_child_weight': 9, 'learning_rate': 0.2075238582103159,
'gamma': 0.4351546631470814, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.
```

```
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
```

```
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:58,912] Trial 36 finished with value: 1.3949053957269564 and
parameters: {'lambda': 0.6270361519430281, 'alpha': 0.6748367996443183,
'subsample': 0.9081901830113768, 'colsample_bytree': 0.27153863998143435,
'max_depth': 7, 'min_child_weight': 8, 'learning_rate': 0.19781530948035064,
'gamma': 0.28529771314142116, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:59,235] Trial 21 finished with value: 1.4285041936767466 and
parameters: {'lambda': 0.7955827268582205, 'alpha': 0.5010030058096901,
'subsample': 0.8688760017291923, 'colsample_bytree': 0.22481540327064328,
'max_depth': 18, 'min_child_weight': 5, 'learning_rate': 0.17768834786338478,
'gamma': 0.40571949392875895, 'n_estimators': 100}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:59,280] Trial 23 finished with value: 1.3701633544507346 and
parameters: {'lambda': 0.4680023540632878, 'alpha': 0.5415704773487606,
'subsample': 0.7788266056562587, 'colsample_bytree': 0.48789090131311597,
'max_depth': 6, 'min_child_weight': 10, 'learning_rate': 0.2321717120332306,
'gamma': 0.45371725142591995, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:26:59,507] Trial 29 finished with value: 1.386959420683364 and
parameters: {'lambda': 0.5583065000917709, 'alpha': 0.4676352884749293,
'subsample': 0.6379080848705194, 'colsample_bytree': 0.7137456381951282,
'max_depth': 7, 'min_child_weight': 11, 'learning_rate': 0.202680476135171,
```

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'gamma': 0.7026310102299749, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:26:59,662] Trial 0 finished with value: 1.3688371828476866 and
parameters: {'lambda': 0.5176692710033693, 'alpha': 0.3002797856871883,
'subsample': 0.4662572669500916, 'colsample_bytree': 0.8469955733287955,
'max_depth': 3, 'min_child_weight': 4, 'learning_rate': 0.1831146822338498,
'gamma': 0.5431380480108643, 'n_estimators': 450}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:00,113] Trial 45 finished with value: 1.4127720809283675 and
parameters: {'lambda': 0.6365519605045087, 'alpha': 0.5604517381236429,
'subsample': 0.6151462304101118, 'colsample_bytree': 0.6165615639652682,
'max_depth': 12, 'min_child_weight': 6, 'learning_rate': 0.2255163164758972,
'gamma': 0.7511117930273243, 'n_estimators': 150}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:00,563] Trial 25 finished with value: 1.3874002866538506 and
parameters: {'lambda': 0.5960126358516148, 'alpha': 0.5996605541457922,
'subsample': 0.4685642566063926, 'colsample_bytree': 0.5767466646285972,
'max_depth': 12, 'min_child_weight': 5, 'learning_rate': 0.20605098940903188,
'gamma': 0.5022421776226014, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:27:00,630] Trial 15 finished with value: 1.3925102823630942 and
parameters: {'lambda': 0.9352664931537937, 'alpha': 0.7881686062037438,
'subsample': 0.29376002559141273, 'colsample_bytree': 0.6026246079405486,
'max_depth': 12, 'min_child_weight': 2, 'learning_rate': 0.17414101493214587,
'gamma': 0.6811739769376742, 'n_estimators': 150}. Best is trial 14 with value:
1.2227390130648852.
```

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[I 2025-10-26 18:27:00,653] Trial 41 finished with value: 1.3598402765685795 and
parameters: {'lambda': 0.5715739513392262, 'alpha': 0.7647238514405665,
'subsample': 0.4571996591688321, 'colsample_bytree': 0.6286847531576615,
'max_depth': 11, 'min_child_weight': 14, 'learning_rate': 0.17462509810048474,
'gamma': 0.520502149868453, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:01,296] Trial 22 finished with value: 1.4208529197326045 and
parameters: {'lambda': 0.6669443494574555, 'alpha': 0.7681019664344899,
'subsample': 0.30104109589336564, 'colsample_bytree': 0.7036673300363057,
'max_depth': 7, 'min_child_weight': 3, 'learning_rate': 0.19964388189374738,
'gamma': 0.11494536191659754, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:27:01,397] Trial 38 finished with value: 1.4158723256069843 and
parameters: {'lambda': 0.6699915996895356, 'alpha': 0.5493747979837502,
'subsample': 0.49622584258191, 'colsample_bytree': 0.9073678769952969,
'max_depth': 7, 'min_child_weight': 11, 'learning_rate': 0.19725423648886953,
'gamma': 0.2031648288075092, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:01,527] Trial 13 finished with value: 1.4470755523764727 and
parameters: {'lambda': 0.6453782823813139, 'alpha': 0.29034853841903374,
'subsample': 0.7122307553071687, 'colsample_bytree': 0.8723790368115882,
'max_depth': 6, 'min_child_weight': 14, 'learning_rate': 0.25218266705749537,
'gamma': 0.08980965891131687, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:03,041] Trial 27 finished with value: 1.3815954563111879 and
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parameters: {'lambda': 0.8383190909037048, 'alpha': 0.5964320836080235,
'subsample': 0.3550296232721068, 'colsample_bytree': 0.6684352420843171,
'max_depth': 12, 'min_child_weight': 12, 'learning_rate': 0.22256488353194104,
'gamma': 0.5911852390927317, 'n_estimators': 350}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:03,419] Trial 31 finished with value: 1.4125571226488265 and
parameters: {'lambda': 0.47523482113734244, 'alpha': 0.6258781827962059,
'subsample': 0.6473157439199074, 'colsample_bytree': 0.6188348357470823,
'max_depth': 8, 'min_child_weight': 12, 'learning_rate': 0.1919768529938689,
'gamma': 0.5960723608892362, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(
/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:03,678] Trial 32 finished with value: 1.4286640499644547 and
parameters: {'lambda': 0.563459098811674, 'alpha': 0.5372997810896898,
'subsample': 0.6130663799234966, 'colsample_bytree': 0.7443720621606547,
'max_depth': 12, 'min_child_weight': 8, 'learning_rate': 0.22710060232988585,
'gamma': 0.7249969407649391, 'n_estimators': 200}. Best is trial 14 with value:
1.2227390130648852.
[I 2025-10-26 18:27:03,866] Trial 24 finished with value: 1.4354136182404158 and
parameters: {'lambda': 0.6650391331305219, 'alpha': 0.33547209847872805,
'subsample': 0.3398693053466418, 'colsample_bytree': 0.6402147872639004,
'max_depth': 9, 'min_child_weight': 9, 'learning_rate': 0.2578722740834018,
'gamma': 0.6157013555470007, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:04,113] Trial 7 finished with value: 1.396980434531381 and
parameters: {'lambda': 0.46745075494057303, 'alpha': 0.40643461340914977,
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'subsample': 0.757538005701941, 'colsample_bytree': 0.459141770267226,
'max_depth': 9, 'min_child_weight': 19, 'learning_rate': 0.21562006421994828,
'gamma': 0.10108009637717275, 'n_estimators': 250}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:04,587] Trial 43 finished with value: 1.4010798250520544 and
parameters: {'lambda': 0.3647010793801451, 'alpha': 0.74753076505203,
'subsample': 0.41750742829011167, 'colsample_bytree': 0.5827645475421628,
'max_depth': 10, 'min_child_weight': 13, 'learning_rate': 0.2148138439002082,
'gamma': 0.8856208143311588, 'n_estimators': 350}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:04,818] Trial 30 finished with value: 1.4147786245199923 and
parameters: {'lambda': 0.6659265908841544, 'alpha': 0.7286802493016418,
'subsample': 0.8683727276995894, 'colsample_bytree': 0.5294308053744293,
'max_depth': 11, 'min_child_weight': 9, 'learning_rate': 0.17795748470732858,
'gamma': 0.26068556883859506, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:05,485] Trial 42 finished with value: 1.392066288449103 and
parameters: {'lambda': 0.6517313744688435, 'alpha': 0.43390977051779084,
'subsample': 0.353054703171008, 'colsample_bytree': 0.6053867795851617,
'max_depth': 7, 'min_child_weight': 11, 'learning_rate': 0.2089457800103165,
'gamma': 0.717043231029463, 'n_estimators': 350}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:05,746] Trial 44 finished with value: 1.381410736349066 and
parameters: {'lambda': 0.40281959251256316, 'alpha': 0.5924276793032748,
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'subsample': 0.8759742403835711, 'colsample_bytree': 0.5704503807247772,
'max_depth': 9, 'min_child_weight': 9, 'learning_rate': 0.1960692776256755,
'gamma': 0.4555199077671972, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:06,281] Trial 8 finished with value: 1.438332999964637 and
parameters: {'lambda': 0.552966018494892, 'alpha': 0.7737154585448853,
'subsample': 0.3562256958077952, 'colsample_bytree': 0.7734818119970333,
'max_depth': 5, 'min_child_weight': 14, 'learning_rate': 0.29718403853566877,
'gamma': 0.913991411012003, 'n_estimators': 500}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:07,712] Trial 39 finished with value: 1.4343038321603043 and
parameters: {'lambda': 0.5331268419362469, 'alpha': 0.6660216519925732,
'subsample': 0.5584559530559046, 'colsample_bytree': 0.30373492544128056,
'max_depth': 14, 'min_child_weight': 8, 'learning_rate': 0.20242363713548123,
'gamma': 0.5425767927529893, 'n_estimators': 300}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:08,297] Trial 20 finished with value: 1.3532254090494564 and
parameters: {'lambda': 0.861801699824434, 'alpha': 0.9732750693888037,
'subsample': 0.8686928290836, 'colsample_bytree': 0.5695004186707171,
'max_depth': 8, 'min_child_weight': 18, 'learning_rate': 0.2523847525190198,
'gamma': 0.5952062449803718, 'n_estimators': 500}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:09,170] Trial 1 finished with value: 1.4374443375968304 and
parameters: {'lambda': 0.33480415459350255, 'alpha': 0.5273499768877405,
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'subsample': 0.3472641406391693, 'colsample_bytree': 0.3407156564683419,
'max_depth': 19, 'min_child_weight': 3, 'learning_rate': 0.21653021804868833,
'gamma': 0.8173127137588244, 'n_estimators': 350}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:11,269] Trial 19 finished with value: 1.490647042200973 and
parameters: {'lambda': 0.4851095478008509, 'alpha': 0.8650010142624731,
'subsample': 0.7911308802782753, 'colsample_bytree': 0.2570353943247623,
'max_depth': 17, 'min_child_weight': 10, 'learning_rate': 0.25133673381037913,
'gamma': 0.03171690773632452, 'n_estimators': 450}. Best is trial 14 with value:
1.2227390130648852.

/home/kunal/anaconda3/lib/python3.10/site-
packages/sklearn/metrics/_regression.py:492: FutureWarning: 'squared' is
deprecated in version 1.4 and will be removed in 1.6. To calculate the root mean
squared error, use the function'root_mean_squared_error'.
  warnings.warn(

[I 2025-10-26 18:27:11,999] Trial 11 finished with value: 1.396280947616608 and
parameters: {'lambda': 0.6930522820832874, 'alpha': 0.7477519445661562,
'subsample': 0.8529556753424867, 'colsample_bytree': 0.9245481758205212,
'max_depth': 11, 'min_child_weight': 1, 'learning_rate': 0.25052794385544486,
'gamma': 0.3380750679325977, 'n_estimators': 450}. Best is trial 14 with value:
1.2227390130648852.

[I 2025-01-29 11:25:20,660] Trial 32 finished with value: 2.5397375785204557 and
parameters: {'lambda': 0.5238877489623508, 'alpha': 0.5869053352875193,
'subsample': 0.7528268887640657, 'colsample_bytree': 0.5443086923654197,
'max_depth': 7, 'min_child_weight': 16, 'learning_rate': 0.25358319474635393,
'gamma': 0.48578454076721433, 'n_estimators': 150}. Best is trial 25 with value:
2.2634835546819767.
[I 2025-01-29 11:25:23,129] Trial 31 finished with value: 2.6802769845284042 and
parameters: {'lambda': 0.5661187878953535, 'alpha': 0.5435252284031926,
'subsample': 0.2581437328922751, 'colsample_bytree': 0.6887766006547653,
'max_depth': 9, 'min_child_weight': 9, 'learning_rate': 0.16951834516324155,
'gamma': 0.0065946824441580065, 'n_estimators': 150}. Best is trial 25 with
value: 2.2634835546819767.
[I 2025-01-29 11:25:31,715] Trial 29 finished with value: 2.351785539257333 and
parameters: {'lambda': 0.6261407372214747, 'alpha': 0.5548604722185964,
'subsample': 0.6592002484901759, 'colsample_bytree': 0.6186188274378228,
'max_depth': 7, 'min_child_weight': 10, 'learning_rate': 0.20644240650913875,
'gamma': 0.6162621842446866, 'n_estimators': 200}. Best is trial 25 with value:
2.2634835546819767.
[I 2025-01-29 11:25:43,766] Trial 17 finished with value: 2.390656791955197 8 and
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parameters: {'lambda': 0.5491560886614726, 'alpha': 0.6409175842337951,
'subsample': 0.4754250865749601, 'colsample_bytree': 0.5962796641617214,
'max_depth': 15, 'min_child_weight': 14, 'learning_rate': 0.19280712890375068,
'gamma': 0.3967283638792617, 'n_estimators': 300}. Best is trial 25 with value:
2.2634835546819767.
[I 2025-01-29 11:25:57,588] Trial 23 finished with value: 2.533442073049291 and
parameters: {'lambda': 0.8694234134882532, 'alpha': 0.36408612607251895,
'subsample': 0.5850716494149868, 'colsample_bytree': 0.4917778269461439,
'max_depth': 10, 'min_child_weight': 9, 'learning_rate': 0.2544417202979866,
'gamma': 0.5909834779653611, 'n_estimators': 350}. Best is trial 25 with value:
2.2634835546819767.
[I 2025-01-29 11:26:02,902] Trial 27 finished with value: 2.217097551610718 and
parameters: {'lambda': 0.5125923815205464, 'alpha': 0.8050078124048075,
'subsample': 0.8056346664900909, 'colsample_bytree': 0.443523848943276,
'max_depth': 11, 'min_child_weight': 9, 'learning_rate': 0.16457765826770449,
'gamma': 0.5117269307750871, 'n_estimators': 250}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:26:12,319] Trial 39 finished with value: 2.577737010525615 and
parameters: {'lambda': 0.6909658075105001, 'alpha': 0.7548030730969641,
'subsample': 0.7230556267735007, 'colsample_bytree': 0.7065573217513961,
'max_depth': 9, 'min_child_weight': 14, 'learning_rate': 0.190164745640063,
'gamma': 0.451455647861999, 'n_estimators': 50}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:26:45,096] Trial 34 finished with value: 2.3307540096015806 and
parameters: {'lambda': 0.6605953663672236, 'alpha': 0.5676931493269126,
'subsample': 0.584590757073878, 'colsample_bytree': 0.658215064127311,
'max_depth': 11, 'min_child_weight': 10, 'learning_rate': 0.18759418013903006,
'gamma': 0.5544965464903634, 'n_estimators': 200}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:27:13,269] Trial 28 finished with value: 2.3983170857896225 and
parameters: {'lambda': 0.6026494331779961, 'alpha': 0.7912383537688643,
'subsample': 0.47498434313154, 'colsample_bytree': 0.7207004367846801,
'max_depth': 12, 'min_child_weight': 10, 'learning_rate': 0.16568164706589775,
'gamma': 0.5414875635565711, 'n_estimators': 250}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:27:57,528] Trial 26 finished with value: 2.2848030371841834 and
parameters: {'lambda': 0.7289985006336019, 'alpha': 0.6425252031708594,
'subsample': 0.5464103621660035, 'colsample_bytree': 0.5148453128541505,
'max_depth': 13, 'min_child_weight': 14, 'learning_rate': 0.21293467000960373,
'gamma': 0.11534400095271269, 'n_estimators': 300}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:28:26,328] Trial 40 finished with value: 2.33471305155213 and
parameters: {'lambda': 0.6079720525589845, 'alpha': 0.5738501214169065,
'subsample': 0.6347363786366595, 'colsample_bytree': 0.4244647877000036,
'max_depth': 11, 'min_child_weight': 8, 'learning_rate': 0.12589036218923433,
'gamma': 0.6931500713725297, 'n_estimators': 200}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:28:27,936] Trial 38 finished with value: 2.3071147154942606 and
```

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parameters: {'lambda': 0.4519654281404446, 'alpha': 0.7284193614083658,
'subsample': 0.811117505177384, 'colsample_bytree': 0.701683331958171,
'max_depth': 17, 'min_child_weight': 10, 'learning_rate': 0.19398827364946608,
'gamma': 0.6587311556146832, 'n_estimators': 100}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:28:28,184] Trial 36 finished with value: 2.4335810167736187 and
parameters: {'lambda': 0.41631641313857637, 'alpha': 0.38869658932750867,
'subsample': 0.44248874807313565, 'colsample_bytree': 0.6471785193381994,
'max_depth': 12, 'min_child_weight': 4, 'learning_rate': 0.2477477577592353,
'gamma': 0.582118809650115, 'n_estimators': 200}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:28:54,217] Trial 41 finished with value: 2.298700699237738 and
parameters: {'lambda': 0.48780953151684175, 'alpha': 0.5816861619502514,
'subsample': 0.6504343373035437, 'colsample_bytree': 0.6942613859884498,
'max_depth': 10, 'min_child_weight': 8, 'learning_rate': 0.17100355500126255,
'gamma': 0.695885290512667, 'n_estimators': 150}. Best is trial 27 with value:
2.217097551610718.
[I 2025-01-29 11:28:59,113] Trial 45 finished with value: 2.2090981554030433 and
parameters: {'lambda': 0.5928370155851933, 'alpha': 0.9252199817752038,
'subsample': 0.7789906661127357, 'colsample_bytree': 0.5230229042859922,
'max_depth': 10, 'min_child_weight': 6, 'learning_rate': 0.16235637881117124,
'gamma': 0.7061449809177778, 'n_estimators': 150}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:28:59,864] Trial 24 finished with value: 2.307052378128035 and
parameters: {'lambda': 0.5197282238292, 'alpha': 0.5437013723594748,
'subsample': 0.5632829627711182, 'colsample_bytree': 0.5528667368013624,
'max_depth': 16, 'min_child_weight': 15, 'learning_rate': 0.10454067783812843,
'gamma': 0.3389046490195651, 'n_estimators': 350}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:00,765] Trial 35 finished with value: 2.221211907853416 and
parameters: {'lambda': 0.6342544331986204, 'alpha': 0.5604950039626901,
'subsample': 0.6049226301870496, 'colsample_bytree': 0.6074745321728268,
'max_depth': 11, 'min_child_weight': 14, 'learning_rate': 0.19022787367193544,
'gamma': 0.5368706345442996, 'n_estimators': 250}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:13,121] Trial 30 finished with value: 2.379938444791064 and
parameters: {'lambda': 0.48982193222395054, 'alpha': 0.6937594729106598,
'subsample': 0.5264925597090919, 'colsample_bytree': 0.8383344729822075,
'max_depth': 11, 'min_child_weight': 10, 'learning_rate': 0.18974594460349115,
'gamma': 0.7736873777019087, 'n_estimators': 300}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:16,901] Trial 46 finished with value: 2.3518147497770827 and
parameters: {'lambda': 0.9205730432529664, 'alpha': 0.5660357445195827,
'subsample': 0.643864696082969, 'colsample_bytree': 0.5434907986781534,
'max_depth': 14, 'min_child_weight': 8, 'learning_rate': 0.13327839630738494,
'gamma': 0.3494444858997893, 'n_estimators': 100}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:21,120] Trial 42 finished with value: 2.21448973334548 and
```

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parameters: {'lambda': 0.5253101330502553, 'alpha': 0.5364655860504517,
'subsample': 0.8301179149853548, 'colsample_bytree': 0.5227004957041375,
'max_depth': 8, 'min_child_weight': 4, 'learning_rate': 0.18896332669212407,
'gamma': 0.7110331563127573, 'n_estimators': 350}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:25,040] Trial 44 finished with value: 2.2250217963486416 and
parameters: {'lambda': 0.5576802034184858, 'alpha': 0.5803221282784556,
'subsample': 0.7976867544360653, 'colsample_bytree': 0.44819620498051654,
'max_depth': 11, 'min_child_weight': 2, 'learning_rate': 0.118083748649538,
'gamma': 0.7758228217759487, 'n_estimators': 250}. Best is trial 45 with value:
2.2090981554030433.

[I 2025-01-29 11:29:33,363] Trial 37 finished with value: 2.357350144179067 and
parameters: {'lambda': 0.69653712456398, 'alpha': 0.6190812493574527,
'subsample': 0.44015170727268443, 'colsample_bytree': 0.8192221906399424,
'max_depth': 11, 'min_child_weight': 16, 'learning_rate': 0.14482920987581802,
'gamma': 0.5458552596207138, 'n_estimators': 300}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:37,853] Trial 43 finished with value: 2.301760869380539 and
parameters: {'lambda': 0.6926935080412153, 'alpha': 0.634419557094194,
'subsample': 0.5696440194387746, 'colsample_bytree': 0.7203124877551217,
'max_depth': 12, 'min_child_weight': 6, 'learning_rate': 0.15794352876583104,
'gamma': 0.6906641113475909, 'n_estimators': 250}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:40,321] Trial 0 finished with value: 2.3835804660200117 and
parameters: {'lambda': 0.3168405634999304, 'alpha': 0.5468963308931032,
'subsample': 0.33941558872170274, 'colsample_bytree': 0.9149315225726338,
'max_depth': 16, 'min_child_weight': 16, 'learning_rate': 0.16468060878311847,
'gamma': 0.1778744252236073, 'n_estimators': 500}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:40,762] Trial 49 finished with value: 2.3766954409600753 and
parameters: {'lambda': 0.6765775399845944, 'alpha': 0.8190285612018389,
'subsample': 0.6951544137159749, 'colsample_bytree': 0.45899821276340624,
'max_depth': 11, 'min_child_weight': 10, 'learning_rate': 0.1719560133968209,
'gamma': 0.7586237842683456, 'n_estimators': 250}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:41,854] Trial 33 finished with value: 2.367404431584162 and
parameters: {'lambda': 0.7149690758096161, 'alpha': 0.5989107829379858,
'subsample': 0.6032747274765593, 'colsample_bytree': 0.8262853951778442,
'max_depth': 14, 'min_child_weight': 15, 'learning_rate': 0.1427975836121181,
'gamma': 0.6554579409533674, 'n_estimators': 300}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:48,076] Trial 2 finished with value: 2.416423240215377 and
parameters: {'lambda': 0.6943735653489058, 'alpha': 0.30657818007667564,
'subsample': 0.35221615558927577, 'colsample_bytree': 0.9532260398413268,
'max_depth': 20, 'min_child_weight': 20, 'learning_rate': 0.14899986054653291,
'gamma': 0.18694013391884745, 'n_estimators': 500}. Best is trial 45 with value:
2.2090981554030433.
```
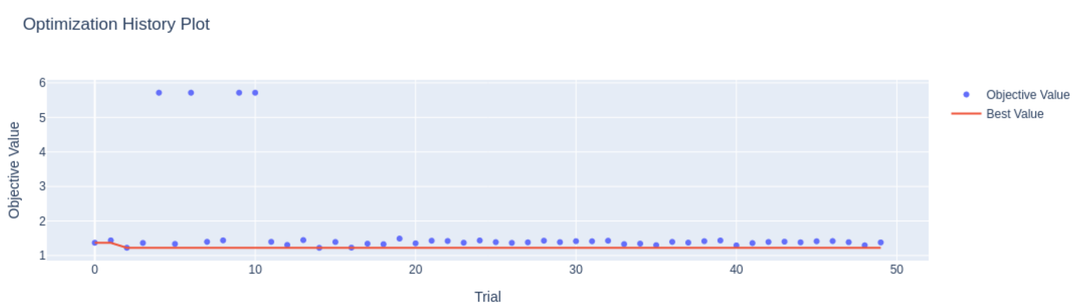

```
[I 2025-01-29 11:29:50,970] Trial 47 finished with value: 2.3647319132668865 and
parameters: {'lambda': 0.6433962398881545, 'alpha': 0.6698834577562622,
'subsample': 0.5671969903436713, 'colsample_bytree': 0.7255489348652293,
'max_depth': 14, 'min_child_weight': 3, 'learning_rate': 0.17073070777732433,
'gamma': 0.6521081870410393, 'n_estimators': 300}. Best is trial 45 with value:
2.2090981554030433.
[I 2025-01-29 11:29:54,727] Trial 48 finished with value: 2.2648677613572032 and
parameters: {'lambda': 0.5906571086101324, 'alpha': 0.7227737729325445,
'subsample': 0.8483898009276571, 'colsample_bytree': 0.794278386051211,
'max_depth': 16, 'min_child_weight': 5, 'learning_rate': 0.1760084552409401,
'gamma': 0.5825590873812354, 'n_estimators': 400}. Best is trial 45 with value:
2.2090981554030433.
```

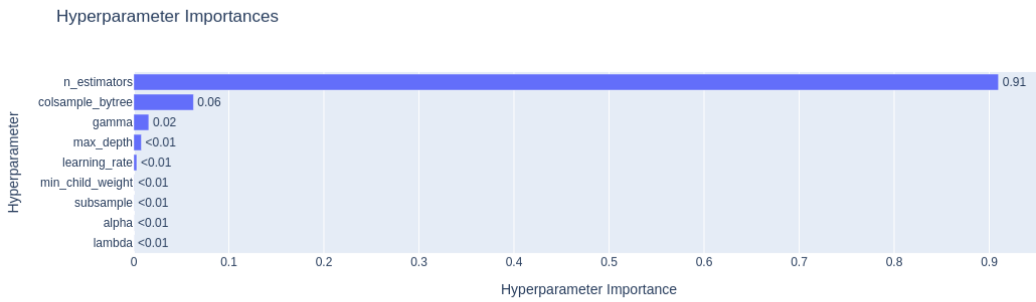
[26]:
```
# import os
# os.getcwd()
study.best_params
```

[26]:
```
{'lambda': 0.6156541051182567,
 'alpha': 0.7262959435632699,
 'subsample': 0.6734027691341069,
 'colsample_bytree': 0.3703731641235166,
 'max_depth': 1,
 'min_child_weight': 19,
 'learning_rate': 0.11647247779004799,
 'gamma': 0.05612923343124632,
 'n_estimators': 200}
```

[27]:
```
fig = optuna.visualization.plot_optimization_history(study)
fig.show()
```



[28]:
```
optuna.visualization.plot_param_importances(study)
```


Hyperparameter Importances
n_estimators
colsample_bytree
gamma
max_depth
learning_rate
min_child_weight
subsample
alpha
lambda
0.91
0.06
0.02
<0.01
<0.01
<0.01
<0.01
<0.01
<0.01
Hyperparameter
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Hyperparameter Importance

```
[ ]:
```